



RESEARCH & INNOVATION UPDATES

Eco-friendly battery as an alternative to lithium



*The low-cost battery made using zinc and lignin, a waste from the paper industry
(Credit: Thor Balkhed/Linköping University)*

Researchers from Linköping University, Karlstad University, and Chalmers University in Sweden have developed an affordable zinc-lignin battery, bypassing the need for lithium. This innovation meets the rising demand for energy storage driven by renewable energy sources like wind and solar. Traditional lithium-ion batteries pose environmental and human rights challenges and are difficult to recycle. The new zinc-based batteries are safer and more sustainable. They overcome previous issues of zinc's reactivity with water by using a water-in-polymer salt electrolyte (WiPSE). These batteries retain charge for a week and maintain 80% performance over 8000 cycles, offering a scalable, environment friendly alternative.

Catalyst slashes iridium use by 95% in hydrogen production

As the world transitions from fossil fuels, hydrogen emerges as a key energy source. Green hydrogen, produced without fossil fuels, faces hurdles due to the scarcity of iridium. The RIKEN Center for Sustainable Resource Science (CSRS) in Japan has made a breakthrough, reducing the required iridium by 95% while maintaining production rates. They developed

a catalyst blend of manganese oxide and iridium, dispersing single iridium atoms across manganese oxide, preventing clumping. This method enabled continuous hydrogen production for over 3000 hours at 82% efficiency. This innovation could scale up hydrogen production, reducing reliance on rare metals and supporting sustainable energy goals.

Autonomous drone using neuromorphic AI

Researchers at Delft University of Technology have developed a drone that flies autonomously using neuromorphic image processing and control, inspired by animal brains. Unlike deep neural networks running on GPUs, neuromorphic processors require less hardware and battery power, making them ideal for small drones. This drone's network processes data 64 times faster and uses three times less energy than a GPU, potentially allowing drones to be as

small and agile as insects or birds. Neuromorphic processors combined with sensors like event-based cameras, which only send signals when pixels change brightness, enhance energy efficiency. The researchers demonstrated a drone using neuromorphic vision and control, with a spiking neural network processing signals from a neuromorphic camera to determine pose and thrust, running on Intel's Loihi chip.